



PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064



**Web Application Development, Deployment
and Security using Linux and cPanel /
HackerHub8 LLP
AN INTERNSHIP REPORT**

Submitted by

Anup Ganiger – 20231CCS3007

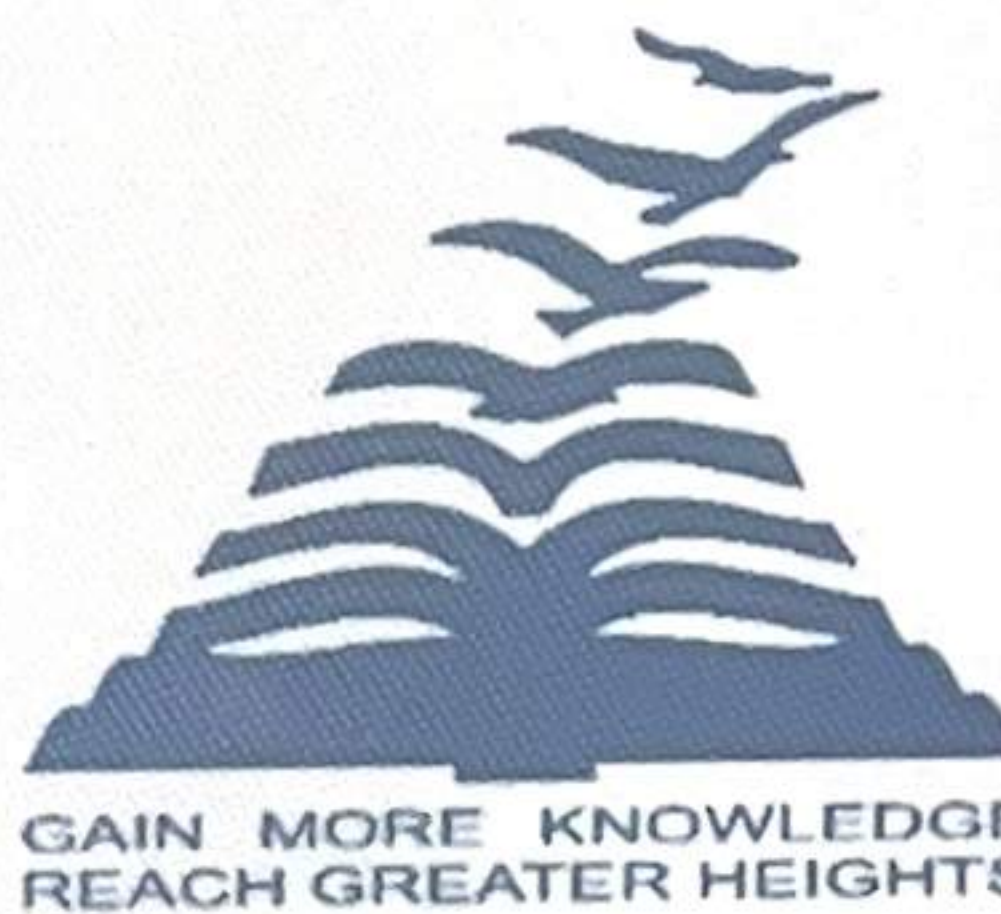
Under the guidance of,

Dr. A.V. Vijaya Kumar

BACHELOR OF TECHNOLOGY

IN

**COMPUTER SCIENCE AND ENGINEERING
(CYBER SECURITY)**



PRESIDENCY UNIVERSITY

BENGALURU

APRIL 2026



PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru - 560064



PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING (SPECIALIZATION DIVISION)

BONAFIDE CERTIFICATE

Certified that this report "Web Application Development, Deployment and Security using Linux and cPanel / HackerHub8 LLP" is a bonafide work of Anup Ganiger (20231CCS3007), who has successfully carried out the internship work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING, CYBER SECURITY during 2025-2026.

Dr. A.V. Vijaya Kumar
Professor
Internship Guide
PSCS
Presidency University

Dr. Geetha A
Associate Professor
Program Internship Coordinator
PSCS
Presidency University

Dr. Geetha A
Associate Professor
School Internship Coordinator
PSCS
Presidency University

Dr. Anandaraj S.P
Professor & Head of the Department
PSCS
Presidency University

Dr. Shakkeera L
Associate Dean
PSCS
Presidency University

Sl.No	Name of the Examiner	Signature
1.	Dr. Man Kumar Gupta	
2.	Dr. Seideni S	



PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064



PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

DECLARATION

I am a student of final year B.Tech in COMPUTER SCIENCE AND ENGINEERING, CYBER SECURITY, at Presidency University, Bengaluru, named **Anup Ganiger**, hereby declare that the internship work titled “**Web Application Development, Deployment and Security using Linux and cPanel / HackerHub8 LLP**” has been independently carried out by me and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE ENGINEERING, CYBER SECURITY during the academic year of 2025-2026. Further, the matter embodied in the internship has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

Anup Ganiger 20231CCS3007

PLACE: BENGALURU

DATE: 22/04/2026

ACKNOWLEDGEMENTS

For completing this internship work, I have received the support and the guidance from many people whom I would like to mention with deep sense of gratitude and indebtedness. I extend my gratitude to our beloved **Chancellor, Pro-Vice Chancellor, and Registrar** for their support and encouragement in completion of the internship.

I would like to sincerely thank my internal guide **Dr A.V. Vijaya Kumar , Professor** Presidency School of Computer Science and Engineering, Presidency University, for his moral support, motivation, timely guidance and encouragement provided to me during the period of internship work.

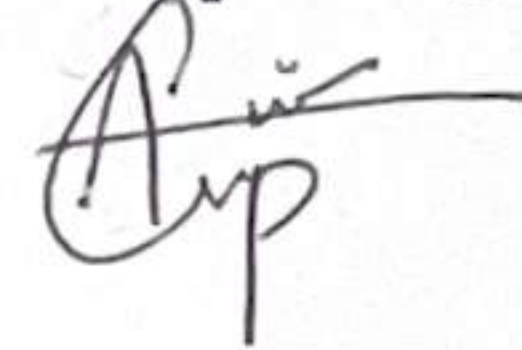
I am also thankful to **Dr. Anandaraj S.P, , Professor, Head of the Department, Presidency School of Computer Science and Engineering** Presidency University, for his mentorship and encouragement.

I express my cordial thanks to **Dr. Shakkeera L**, Associate Dean, Presidency School of Computer Science and Engineering and the Management of Presidency University for providing the required facilities and intellectually stimulating environment that aided in the completion of my internship work.

I am grateful to **Dr. Geetha A**, School Internship Coordinator and Program Internship Coordinator, Presidency School of Computer Science and Engineering, for facilitating problem statements, coordinating reviews, monitoring progress, and providing their valuable support and guidance.

I am also grateful to Teaching and Non-Teaching staff of Presidency School of Computer Science and Engineering and also staff from other departments who have extended their valuable help and cooperation.

Anup Ganiger



INTERNSHIP COMPLETION CERTIFICATE



HACKERHUB8 LLP

Built with Purpose. Secured with Passion. Inspired by Innovation

ISO 9001:2015 | 27001:2022 Certified

9606969233, 8277269695

support@hackerhub8.in

www.hackerhub8.in

Bengaluru, Karnataka, India

INTERNSHIP COMPLETION LETTER

To Whom It May Concern,

This is to certify that **Anup Ganiger** (ID: HH8-7130846) from **Presidency University** has successfully completed the internship at **HackerHub8** from **04/01/2026 to 01/05/2026**.

During this period, the intern worked on the project titled "**Web Application Development, Deployment and Security using Linux and cPanel / HackerHub8 LLP**", focusing on cybersecurity concepts, implementation, and research. The intern was actively involved in project development, documentation, and analysis, including preparation of case studies, research reports, and a white paper.

The performance of **Anup Ganiger** was found to be satisfactory, demonstrating strong technical skills, dedication, and a professional approach towards assigned responsibilities.

We appreciate their contribution and wish them continued success in their future academic and professional endeavors.

For HackerHub8

Ravi G

Director HackerHub8 LLP



Date: 25/04/2026

ACKNOWLEDGEMENTS

For completing this internship work, I have received the support and the guidance from many people whom I would like to mention with deep sense of gratitude and indebtedness. I extend my gratitude to our beloved **Chancellor, Pro-Vice Chancellor, and Registrar** for their support and encouragement in completion of the internship.

I would like to sincerely thank my internal guide **Dr A.V. Vijaya Kumar , Professor** Presidency School of Computer Science and Engineering, Presidency University, for his moral support, motivation, timely guidance and encouragement provided to me during the period of internship work.

I am also thankful to **Dr. Anandaraj S.P. , Professor, Head of the Department, Presidency School of Computer Science and Engineering** Presidency University, for his mentorship and encouragement.

I express my cordial thanks to **Dr. Shakkeera L**, Associate Dean, Presidency School of Computer Science and Engineering and the Management of Presidency University for providing the required facilities and intellectually stimulating environment that aided in the completion of my internship work.

I am grateful to **Dr. Geetha A**, School Internship Coordinator and Program Internship Coordinator, Presidency School of Computer Science and Engineering, for facilitating problem statements, coordinating reviews, monitoring progress, and providing their valuable support and guidance.

I am also grateful to Teaching and Non-Teaching staff of Presidency School of Computer Science and Engineering and also staff from other departments who have extended their valuable help and cooperation.

Anup Ganiger

ABSTRACT

The rapid growth of web technologies and digital platforms has increased the demand for secure and efficiently managed web applications. Organizations today require not only functional websites but also secure, monitored, and well-maintained hosting environments. This internship was carried out to gain practical exposure to real-world web development, deployment, and cybersecurity practices in a Linux-based environment. During the internship, a complete web application was developed and deployed using a Linux operating system with cPanel for hosting and server management. The process included website design and development, domain and hosting configuration, file management, and server-side setup. In addition to deployment, monitoring tools such as AWStats and Webalizer were configured to analyze web traffic, user behavior, and server performance. Furthermore, a Vulnerability Assessment and Penetration Testing (VAPT) process was conducted to identify potential security weaknesses in the web application. The identified vulnerabilities were analyzed and documented along with appropriate mitigation strategies. The results of the internship demonstrate successful end-to-end implementation of a web application lifecycle, from development to secure deployment and monitoring. The VAPT analysis helped in identifying critical security gaps and improving the overall security posture of the system. The use of monitoring tools provided valuable insights into website performance and user interaction. This internship enhanced practical knowledge in web development, Linux system administration, hosting management, and cybersecurity, thereby preparing for real-world industry challenges.

TABLE OF CONTENTS

Sl. No.	Title	Page No.
	Declaration	iii
	Acknowledgement	v
	Abstract	vi
	List of Figures	ix
	List of Tables	x
	Abbreviations	xi
1.	Introduction	
	1.1 Background	1
	1.2 Statistics of Internship	1
	1.3 Prior existing technologies	3
	1.4 Proposed approach	4
	1.5 Objectives	5
	1.6 SDGs	5
	1.7 Overview of internship report	6
2.	Literature Review	7-8
3.	Methodology	8-10
4.	Project Management	
	4.1 Internship timeline	11
	4.2 Risk analysis	12
	4.3 Internship budget	13
5.	Analysis and Design	
	5.1 Requirements	13-14
	5.2 Block Diagram	14-15
	5.3 System Flow Chart	16
	5.4 Domain model specification	17
	5.5 Communication model	18
	5.6 Functional view	19-20
6.	Hardware, Software and Simulation	
	6.1 Hardware	21
	6.2 Software development tools	22-23

	6.3 Software code	24
	6.4 Simulation	25-34
7.	Evaluation and Results	
	7.1 Test points	35
	7.2 Test plan	36
	7.3 Test result	36
	7.4 Insights	37
8.	Social, Legal, Ethical, Sustainability and Safety Aspects	
	8.1 Social aspects	37
	8.2 Legal aspects	38
	8.3 Ethical aspects	38
	8.4 Sustainability aspects	39
	8.5 Safety aspects	39
9.	Conclusion	40
	References	41
	Appendix	42

LIST OF FIGURES

Sl. No.	Figure Name	Caption	Page No.
1	Figure 5.1	Block Diagram	15
2	Figure 5.2	Flow Chart	16
3	Figure 5.3	Conceptual Representation	18
4	Figure 5.4	Functional Workflow	21
5	Figure 6.1	Code Diagram	24
6	Figure 6.2	Cpanel Server Login Page	25
7	Figure 6.3	Cpanel Dashboard	25
8	Figure 6.4	Server Configuration	26
9	Figure 6.5	Server File Manager	26
10	Figure 6.6	Server Usage & Limit	26
11	Figure 6.7	phpMyAdmin Usage Dashboard	27
12	Figure 6.8	phpMyAdmin Usage Dashboard	27
13	Figure 6.9	Server Disk usage	28
14	Figure 6.10	Server Usage Statics	28
15	Figure 6.11	Server Resource Usage	29
16	Figure 6.12	Server DNS Configuration	30
17	Figure 6.13	Server Mail Deliverability	31
18	Figure 6.14	AwStat Dashboard	32
19	Figure 6.15	AwStat Dashboard	33
20	Figure 6.16	Server Bandwidth usage	34

LIST OF TABLES

Sl. No.	Table Name	Table Caption	Page No.
1	Table 7.1	Test Plan	36

ABBREVIATIONS

Abbreviation	Full Form
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
PHP	Hypertext Preprocessor
OS	Operating System
UI	User Interface
URL	Uniform Resource Locator
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
TCP/IP	Transmission Control Protocol / Internet Protocol
DB	Database
SQL	Structured Query Language
MySQL	My Structured Query Language
VAPT	Vulnerability Assessment and Penetration Testing
OWASP	Open Web Application Security Project
cPanel	Control Panel
AWStats	Advanced Web Statistics
Webalizer	Web Log Analyzer
DNS	Domain Name System
FTP	File Transfer Protocol
SDGs	Sustainable Development Goals
VPN	Virtual Private Network
XML	Extensible Markup Language

Chapter 1

INTRODUCTION

1.1 Background

In the modern digital era, web applications play a critical role in enabling businesses, organizations, and individuals to establish an online presence and deliver services efficiently. With the increasing dependency on web-based platforms, there is a growing need for secure, scalable, and well-managed web infrastructures. However, many web applications are deployed without proper security configurations, monitoring mechanisms, and performance optimization, making them vulnerable to cyber threats.

The field of cybersecurity has therefore become essential in ensuring the confidentiality, integrity, and availability of web applications. Alongside development, secure deployment practices and continuous monitoring are key components in maintaining a robust web environment. Technologies such as Linux-based servers and control panels like cPanel are widely used in the industry for hosting and managing web applications due to their flexibility and efficiency.

This internship was designed to provide hands-on experience in the complete lifecycle of a web application, including development, deployment, monitoring, and security assessment. The project focused on building a fully functional website, configuring it on a Linux server using cPanel, and implementing monitoring tools such as AWStats and Webalizer to track performance and user activity.

In addition, a Vulnerability Assessment and Penetration Testing (VAPT) process was carried out to identify and analyze potential security risks in the deployed application. This practical exposure helped in understanding real-world challenges in web security, server management, and system monitoring, thereby bridging the gap between theoretical knowledge and industry practices.

1.2 Statistics of Internship

The internship was structured to provide practical exposure to web development, deployment, monitoring, and cybersecurity practices. The following statistics summarize the key activities and outcomes achieved during the internship period:

- **Duration of Internship:** [e.g., 8 Weeks / 2 Months]
- **Working Hours:** Approximately [e.g., 4–6 hours per day]
- **Total Project Completed:** 1 (Full Web Application Development & Deployment)
- **Technologies Used:**
 - Linux Operating System
 - cPanel Hosting Environment
 - HTML, CSS, JavaScript (Frontend Development)
 - PHP / Backend Technologies (if used)
 - MySQL Database (if used)
- **Development Work:**
 - Designed and developed a complete functional website
 - Configured domain and hosting environment
 - Managed file systems and server-side setup
- **Deployment Activities:**
 - Website successfully hosted on a Linux server
 - cPanel configured for file management, databases, and domains
 - Server environment optimized for performance
- **Monitoring & Analytics:**
 - AWStats configured for traffic analysis
 - Webalizer configured for log-based analytics
 - Monitoring of user visits, bandwidth usage, and access patterns
- **Security Assessment:**
 - Conducted Vulnerability Assessment and Penetration Testing (VAPT)
 - Identified common web vulnerabilities
 - Prepared detailed VAPT report with mitigation strategies
- **Tools & Utilities Used:**
 - cPanel
 - Linux Terminal
 - Web Analytics Tools (AWStats, Webalizer)
 - Security Testing Tools (e.g., Nmap, basic scanning tools if used)
- **Key Outcomes:**
 - Successful end-to-end website deployment
 - Implementation of monitoring systems
 - Identification and mitigation of security vulnerabilities
 - Improved understanding of real-world cybersecurity practices

1.3 Prior Existing Technologies

Before the implementation of the proposed system in this internship, several traditional technologies and approaches were commonly used for web application development, deployment, and monitoring. These existing systems provided basic functionality but often lacked integration, security, and efficient monitoring capabilities.

In terms of web development, static websites built using HTML, CSS, and basic JavaScript were widely used. These websites were simple and easy to deploy but lacked dynamic features and scalability. For more advanced applications, server-side technologies such as PHP and MySQL were used; however, they were often implemented without standardized security practices.

For deployment, traditional hosting methods relied on manual server configuration or shared hosting environments without proper optimization. Although control panels like cPanel were available, they were not always fully utilized for efficient server management, leading to misconfigurations and performance issues.

Monitoring of web applications was limited or sometimes completely absent. Basic log files were generated by servers, but tools like AWStats and Webalizer were not consistently configured or analyzed. As a result, administrators had limited visibility into website traffic, user behavior, and system performance.

From a security perspective, many systems were deployed without proper vulnerability assessment. Security testing such as Vulnerability Assessment and Penetration Testing (VAPT) was often ignored or conducted only after issues arose. This resulted in common vulnerabilities such as weak authentication, misconfigured servers, and exposure to web-based attacks.

Overall, the prior existing technologies provided foundational capabilities but lacked an integrated approach combining development, secure deployment, monitoring, and proactive security assessment. This gap highlighted the need for a more structured and security-focused solution, which is addressed in the proposed approach of this internship.

1.4 Proposed Approach

To overcome the limitations of traditional web development and deployment practices, this internship adopted an integrated and security-focused approach that combines web application development, structured deployment, continuous monitoring, and vulnerability assessment.

The proposed approach began with the development of a complete web application using modern web technologies. The website was designed to be functional, user-friendly, and structured for real-world deployment. Following development, the application was deployed on a Linux-based server environment using cPanel, which enabled efficient management of files, domains, databases, and server configurations.

Unlike traditional methods, this approach emphasized proper configuration and optimization of the hosting environment. The deployment process included setting up the domain, managing server directories, and ensuring that the application was accessible and stable in a live environment.

To enhance visibility and performance tracking, monitoring tools such as AWStats and Webalizer were configured. These tools provided detailed insights into website traffic, visitor behavior, bandwidth usage, and server logs, enabling better analysis and decision-making.

A key aspect of the proposed approach was the inclusion of security assessment through Vulnerability Assessment and Penetration Testing (VAPT). The web application was tested for common vulnerabilities, and the findings were documented in a structured VAPT report. Necessary recommendations and mitigation strategies were identified to improve the security posture of the system.

Overall, the proposed approach integrates development, deployment, monitoring, and security into a single workflow. This ensures that the web application is not only functional but also secure, optimized, and continuously monitored, aligning with industry best practices.

1.5 Objectives

The primary objectives of this internship were to gain practical knowledge and hands-on experience in web development, deployment, monitoring, and cybersecurity practices. The specific objectives are as follows:

- To design and develop a fully functional web application using modern web technologies
- To understand and implement web hosting using a Linux-based environment
- To configure and manage hosting services using cPanel
- To deploy the developed website on a live server environment
- To monitor website performance and user activity using tools such as AWStats and Webalizer
- To perform Vulnerability Assessment and Penetration Testing (VAPT) on the web application
- To identify security vulnerabilities and suggest appropriate mitigation strategies
- To gain knowledge of server management, file handling, and domain configuration
- To understand real-world cybersecurity practices and implementation
- To enhance problem-solving skills and technical expertise in web security and deployment

These objectives guided the overall workflow of the internship and ensured a comprehensive learning experience across development, deployment, monitoring, and security domains.

1.6 Sustainable Development Goals (SDGs)

The work carried out during this internship aligns with several United Nations Sustainable Development Goals (SDGs), particularly in the areas of innovation, infrastructure, and secure digital systems.

- **SDG 9: Industry, Innovation, and Infrastructure**
The development and deployment of a web application using modern technologies contribute to building reliable and sustainable digital infrastructure. The use of Linux-based servers and structured hosting environments promotes innovation and efficient system design.
- **SDG 16: Peace, Justice, and Strong Institutions**
The implementation of cybersecurity practices, including Vulnerability Assessment and Penetration Testing (VAPT), helps in strengthening the security of digital platforms. Secure web applications are essential for protecting user data and ensuring trust in online systems.
- **SDG 8: Decent Work and Economic Growth**
By gaining practical skills in web development, deployment, and cybersecurity, the internship enhances employability and prepares individuals for professional roles in the technology sector, thereby contributing to economic growth.

Overall, this internship supports the development of secure, reliable, and innovative digital solutions while promoting skill development and responsible use of technology in alignment with global sustainability goals.

1.7 Overview of Internship Report

This internship report is organized into multiple chapters, each describing different aspects of the work carried out during the internship.

- **Chapter 1: Introduction**

This chapter provides the background, objectives, existing technologies, proposed approach, and alignment with Sustainable Development Goals (SDGs).

- **Chapter 2: Literature Review**

This chapter discusses existing research, tools, and technologies related to web development, hosting environments, monitoring systems, and cybersecurity practices.

- **Chapter 3: Methodology**

This chapter explains the step-by-step process followed during the internship, including development, deployment, monitoring, and security assessment.

- **Chapter 4: Project Management**

This chapter outlines the internship timeline, risk analysis, and resource or budget considerations.

- **Chapter 5: Analysis and Design**

This chapter describes system requirements, design structure, flowcharts, and overall architecture of the developed solution.

- **Chapter 6: Hardware, Software and Simulation**

This chapter details the tools, technologies, software platforms, and any simulation or testing environments used during the project.

- **Chapter 7: Evaluation and Results**

This chapter presents the testing procedures, results obtained, and insights derived from the implementation and security analysis.

- **Chapter 8: Social, Legal, Ethical, Sustainability and Safety Aspects**

This chapter discusses the broader impact of the project, including ethical considerations, legal compliance, and sustainability aspects.

- **Chapter 9: Conclusion**

This chapter summarizes the overall work, key learnings, and outcomes of the internship.

- **References and Appendix**

These sections include sources of information, supporting documents, and additional data related to the project.

This structured organization ensures a clear and comprehensive presentation of the internship work and outcomes.

Chapter 2

LITERATURE SURVEY

The development, deployment, and security of web applications have been widely studied in the fields of web engineering and cybersecurity. Various technologies and methodologies have been proposed to improve performance, scalability, and security of web-based systems.

Web development has evolved from static HTML-based pages to dynamic and interactive applications using technologies such as CSS, JavaScript, and server-side scripting languages like PHP. Modern web applications are designed to be responsive, user-friendly, and scalable, supporting a wide range of devices and users. Research highlights the importance of structured development practices to ensure maintainability and performance (Pressman, *Software Engineering: A Practitioner's Approach*, 2014; Murugesan, *Web Engineering: Managing Diversity and Complexity of Web Application Development*, 2001).

In terms of deployment, Linux-based servers are widely preferred due to their stability, flexibility, and open-source nature. Control panels such as cPanel have been extensively used for simplifying server management tasks, including domain configuration, file handling, database management, and email services. Studies indicate that proper server configuration plays a crucial role in ensuring system reliability and uptime (Nemeth et al., *UNIX and Linux System Administration Handbook*, 2017; Peltier, *Information Security Policies, Procedures, and Standards*, 2016).

Monitoring tools such as AWStats and Webalizer are commonly used for analyzing web traffic and server logs. These tools provide insights into user behavior, page visits, bandwidth consumption, and access patterns. Literature suggests that continuous monitoring is essential for performance optimization and early detection of anomalies in web applications (Behl & Behl, *Cybersecurity and Cyberwar: What Everyone Needs to Know*, 2017).

Cybersecurity is a critical aspect of web application management. Vulnerability Assessment and Penetration Testing (VAPT) are widely adopted practices for identifying and mitigating security risks. Research shows that common web vulnerabilities, such as misconfigurations, weak authentication, and input validation issues, can lead to serious security breaches if not properly addressed. Implementing regular security testing helps in strengthening the overall security posture of web systems (Stuttard & Pinto, *The Web Application Hacker's Handbook*, 2011; OWASP, *OWASP Top 10 Web Application Security Risks*, 2021).

Overall, existing literature emphasizes the need for an integrated approach that combines development, deployment, monitoring, and security. This internship aligns with these findings by implementing a complete workflow that ensures not only functionality but also security and performance of the web application.

Chapter 3

METHODOLOGY

The methodology adopted during this internship follows a structured and systematic approach to ensure the successful development, deployment, monitoring, and security assessment of a web application. The process is divided into multiple phases, each focusing on a specific aspect of the project lifecycle.

3.1 Requirement Analysis

The initial phase involved understanding the project requirements and defining the scope of work. The key requirements included:

- Development of a functional and responsive web application
- Deployment on a Linux-based hosting environment
- Implementation of monitoring tools for performance tracking
- Conducting security testing using VAPT methodologies

This phase helped in identifying system needs, tools, and expected outcomes.

3.2 System Design and Planning

A structured design approach was followed before implementation. The website architecture, navigation flow, and content structure were planned. Decisions regarding hosting environment, domain configuration, and security considerations were also made in this stage.

3.3 Web Application Development

The development phase focused on building a complete and user-friendly website:

- Frontend development using HTML, CSS, and JavaScript
- Implementation of responsive design for better user experience
- Integration of backend components (if applicable)
- Ensuring proper file structuring and code organization

The application was developed with attention to usability, performance, and maintainability.

3.4 Linux Server Environment Setup

A Linux-based server environment was used for hosting the application. This phase included:

- Understanding Linux file system structure

- Managing permissions and user access
- Basic command-line operations for server management
- Preparing the environment for web hosting

This ensured a stable and secure hosting foundation.

3.5 Deployment using cPanel

The deployment process was carried out using cPanel, which simplified server management tasks:

- Domain and DNS configuration
- Uploading website files via File Manager
- Database creation and configuration (if required)
- Email and hosting settings management
- Ensuring proper file permissions and directory structure

The website was successfully made live and accessible over the internet.

3.6 Monitoring and Log Analysis

To track website activity and performance, monitoring tools were implemented:

- **AWStats:** Provided detailed reports on visitors, traffic sources, and bandwidth usage
- **Webalizer:** Offered graphical representation of web server logs

Log analysis helped in understanding user behavior, peak traffic times, and potential anomalies in system usage.

3.7 Security Testing (VAPT)

A critical phase of the methodology involved performing Vulnerability Assessment and Penetration Testing:

- Identification of common vulnerabilities such as misconfigurations, weak authentication, and exposure issues
- Use of basic security scanning tools and manual testing techniques
- Analysis of server and application-level security
- Documentation of vulnerabilities with severity levels
- Recommendation of mitigation strategies

This phase enhanced the overall security of the deployed application.

3.8 Performance and Functional Testing

The deployed system was tested to ensure:

- Proper loading of web pages
- Server responsiveness and uptime
- Correct functioning of all features
- Absence of critical errors

Testing ensured that the application met functional and performance requirements.

3.9 Documentation and Reporting

All activities carried out during the internship were documented in a structured manner:

- Development process documentation
- Deployment steps and configurations
- Monitoring setup details
- VAPT findings and recommendations

This documentation forms the basis of the final internship report.

3.10 Workflow Summary

The overall workflow followed a sequential approach:

Requirement Analysis → Design → Development → Deployment → Monitoring →
Security Testing → Evaluation → Documentation

This comprehensive methodology ensured that the project was completed efficiently, covering all essential aspects of web application development, deployment, monitoring, and cybersecurity.

Chapter 4

PROJECT MANAGEMENT

4.1 Internship timeline

The internship was carried out over a structured period, with each phase focusing on specific tasks related to web development, deployment, monitoring, and security assessment. The timeline of activities is as follows:

- **Week 1: Orientation and Requirement Analysis**
 - Introduction to internship objectives and tools
 - Understanding project requirements and scope
 - Basic overview of Linux environment and hosting concepts
- **Week 2: Planning and Design**
 - Designing website structure and layout
 - Planning navigation flow and content organization
 - Selecting technologies and tools for development
- **Week 3–4: Web Application Development**
 - Developing frontend using HTML, CSS, and JavaScript
 - Implementing responsive design
 - Structuring files and organizing code
 - Testing basic functionality
- **Week 5: Server Setup and Deployment**
 - Setting up Linux-based hosting environment
 - Configuring cPanel and domain settings
 - Uploading website files to server
 - Ensuring website accessibility online
- **Week 6: Monitoring and Analytics Setup**
 - Configuring AWStats for traffic analysis
 - Setting up Webalizer for log monitoring
 - Analyzing initial traffic and server data
- **Week 7: Security Testing (VAPT)**
 - Performing vulnerability assessment
 - Identifying security issues in the application
 - Documenting findings and risk levels
 - Suggesting mitigation strategies
- **Week 8: Testing, Documentation, and Final Reporting**
 - Final system testing and validation
 - Preparing VAPT report and project documentation
 - Compiling internship report
 - Reviewing overall project outcomes

4.2 Risk analysis

Risk analysis is an essential part of project management, especially in web development and cybersecurity. During this internship, several potential risks were identified across development, deployment, monitoring, and security phases. Appropriate mitigation strategies were considered to minimize their impact.

- **Technical Risks**
 - *Risk:* Errors in website development, broken functionality, or code issues
 - *Mitigation:* Regular testing, debugging, and validation of features during development
- **Server Configuration Risks**
 - *Risk:* Misconfiguration of Linux server or cPanel settings leading to downtime or vulnerabilities
 - *Mitigation:* Proper configuration practices, verification of settings, and controlled access management
- **Security Risks**
 - *Risk:* Presence of vulnerabilities such as weak authentication, improper permissions, or exposed data
 - *Mitigation:* Conducting Vulnerability Assessment and Penetration Testing (VAPT), applying security best practices, and implementing recommended fixes
- **Deployment Risks**
 - *Risk:* Failure in website deployment, incorrect file paths, or domain configuration issues
 - *Mitigation:* Step-by-step deployment process, validation after deployment, and backup of files
- **Data Loss Risks**
 - *Risk:* Loss of website data due to server errors or accidental deletion
 - *Mitigation:* Maintaining backups and using secure storage practices
- **Performance Risks**
 - *Risk:* Slow website performance or server overload
 - *Mitigation:* Monitoring using AWStats and Webalizer, optimizing files and server usage
- **Monitoring Risks**
 - *Risk:* Incorrect or incomplete log analysis leading to missed insights
 - *Mitigation:* Proper configuration of monitoring tools and regular review of logs
- **Time Management Risks**
 - *Risk:* Delay in completing tasks within internship duration
 - *Mitigation:* Following a structured timeline and prioritizing tasks

4.3 Internship budget

The internship project was carried out with minimal financial requirements, as most of the tools and technologies used were open-source or freely available. The budget primarily included costs related to hosting services and basic infrastructure.

The estimated budget is as follows:

- **Domain Registration:**
Cost incurred for registering a domain name for hosting the website (if applicable).
- **Web Hosting / cPanel Access:**
Charges associated with hosting the website on a Linux server using cPanel (may be provided by the organization or purchased).
- **Internet and Connectivity:**
Cost of internet usage for development, deployment, and testing activities.
- **Software and Tools:**
Most tools used, such as Linux OS, development technologies (HTML, CSS, JavaScript), and monitoring tools (AWStats, Webalizer), were open-source and did not incur additional cost.
- **Security Testing Tools:**
Basic vulnerability assessment tools used were either open-source or freely available.
- **Miscellaneous Expenses:**
Includes minor costs such as documentation, electricity, and system usage.

Chapter 5

ANALYSIS AND DESIGN

5.1 Requirements

The requirements of the system were identified to ensure the successful development, deployment, monitoring, and security of the web application. These requirements are categorized into functional and non-functional requirements.

5.1.1 Functional Requirements

Functional requirements define the core features and operations of the system:

- The system should allow development of a fully functional web application
- The website should be accessible through a web browser over the internet
- The system should support deployment on a Linux-based server
- cPanel should be used for managing hosting services, files, and domains
- The system should enable uploading, editing, and managing website files
- The system should support database integration (if applicable)
- Monitoring tools such as AWStats and Webalizer should be configured
- The system should generate and display website traffic and usage reports
- The system should support security testing through VAPT
- The system should allow identification and reporting of vulnerabilities

5.1.2 Non-Functional Requirements

Non-functional requirements define the performance and quality attributes of the system:

- Performance:
The website should load efficiently and handle user requests with minimal delay
- Security:
The system should be protected against common vulnerabilities through proper configurations and testing
- Reliability:
The web application should be consistently available with minimal downtime
- Scalability:
The system should be capable of handling increased traffic and future expansion
- Usability:
The website should have a user-friendly interface and easy navigation
- Maintainability:
The system should be easy to update, manage, and troubleshoot
- Compatibility:
The website should work across different browsers and devices
- Availability:
The hosted application should be accessible at all times through proper server management

These requirements ensured that the system was not only functional but also secure, reliable, and efficient in real-world conditions.

5.2 Block Diagram

The block diagram represents the overall architecture of the system, showing how different components interact with each other during web application development, deployment, monitoring, and security testing.

Block Diagram Description

The system consists of the following main components:

1. **User (Client Side)**
The end user accesses the website through a web browser. User requests are sent to the web server over the internet.
2. **Web Application (Frontend + Backend)**
The developed website processes user requests and provides appropriate responses. It includes frontend (UI) and backend logic (if implemented).
3. **Web Server (Linux + cPanel)**
The web server hosts the application and manages requests. cPanel is used to manage files, domains, databases, and server configurations.
4. **Database (Optional)**
Stores application data such as user information, content, or logs (if used in the project).
5. **Monitoring Tools (AWStats & Webalizer)**
These tools analyze server logs and provide insights into traffic, user behavior, and performance.
6. **Security Testing Module (VAPT)**
Performs vulnerability assessment and penetration testing to identify and analyze security issues.

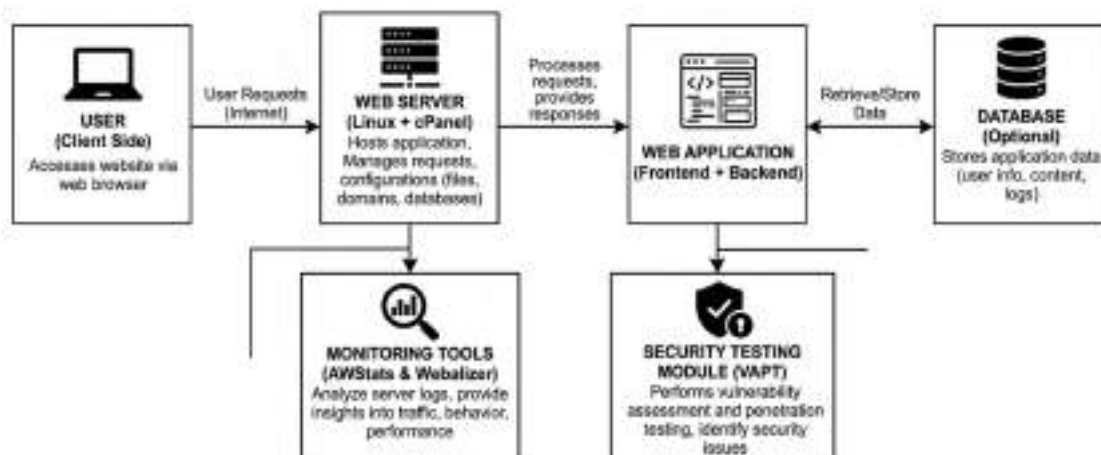


Figure 5.1

Block Diagram Flow

User → Internet → Web Server (Linux + cPanel) → Web Application → Database
↓
Monitoring Tools (AWStats, Webalizer)
↓
Security Testing (VAPT)

Explanation

The user interacts with the web application through a browser. The request is processed by the web server, which hosts the application. If required, the application communicates with the database to retrieve or store data. Monitoring tools continuously analyze server logs to track performance and usage. Simultaneously, security testing is performed to detect vulnerabilities and improve system security.

This block diagram illustrates the integrated workflow of development, deployment, monitoring, and security in the system.

5.3 System Flow Chart

The system flow chart represents the step-by-step workflow of the web application from user interaction to deployment, monitoring, and security testing. It illustrates how data and processes flow within the system.

Flow Description

The process begins when a user accesses the website through a web browser. The request is sent over the internet to the web server hosted on a Linux environment using cPanel. The server processes the request and forwards it to the web application.

The web application handles the request and, if necessary, interacts with the database to retrieve or store data. The processed response is then sent back to the user through the web server.

Simultaneously, server logs are generated and analyzed using monitoring tools such as AWStats and Webalizer. These tools provide insights into traffic, usage patterns, and performance.

In parallel, security testing is conducted through Vulnerability Assessment and Penetration Testing (VAPT) to identify and mitigate potential vulnerabilities in the system.

Flowchart Diagram

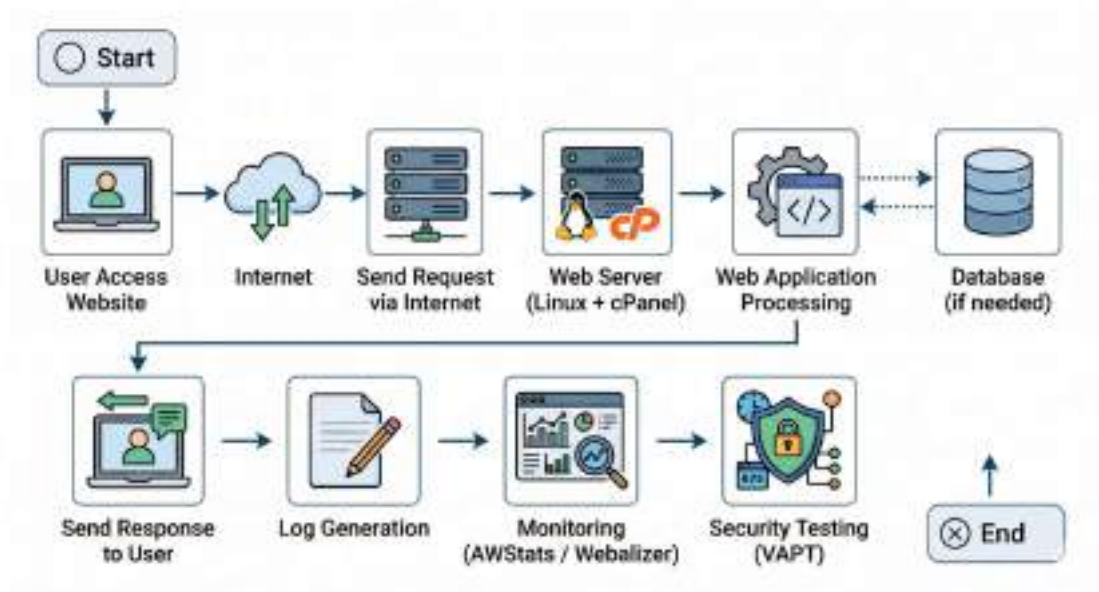


Figure 5.2

5.4 Domain model specification

The domain model represents the key entities involved in the system and their relationships. It provides a conceptual view of how different components of the web application interact within the system.

Domain Entities

The main entities identified in the system are:

- **User**
Represents the end user who accesses the website through a browser. The user interacts with the web application by sending requests and receiving responses.
- **Web Application**
The core component that processes user requests and generates responses. It includes frontend and backend logic.

- **Web Server (Linux + cPanel)**
Hosts the web application and manages incoming requests. It handles file storage, domain management, and server configurations.
- **Database (Optional)**
Stores application data such as user information, content, and logs (if applicable).
- **Monitoring System (AWStats & Webalizer)**
Collects and analyzes server logs to provide insights into website traffic, usage, and performance.
- **Security Module (VAPT)**
Responsible for identifying vulnerabilities and assessing the security of the web application.

Relationships Between Entities

- The **User** interacts with the **Web Application** through the internet
- The **Web Application** is hosted on the **Web Server**
- The **Web Application** communicates with the **Database** to store and retrieve data
- The **Web Server** generates logs that are analyzed by the **Monitoring System**
- The **Security Module (VAPT)** interacts with the **Web Application** and **Web Server** to identify vulnerabilities

Conceptual Representation



Figure 5.3

Explanation

The domain model highlights how the system components are logically connected. The user initiates interaction, which is processed by the web application hosted on the server. The database supports data storage operations, while monitoring tools continuously analyze system activity. The security module ensures that the system remains protected against vulnerabilities.

This domain model provides a clear understanding of system structure and helps in designing, analyzing, and improving the overall architecture.

5.5 Communication Model

The communication model defines how data is transmitted between different components of the system, including the user, web server, web application, database, monitoring tools, and security modules. It ensures efficient and secure exchange of information within the system.

Communication Components

- **Client (User Browser):**
Sends requests and receives responses using web protocols
- **Web Server (Linux + cPanel):**
Acts as an intermediary that receives client requests and forwards them to the web application
- **Web Application:**
Processes requests and communicates with the database if required
- **Database (Optional):**
Stores and retrieves data based on application requests
- **Monitoring Tools (AWStats & Webalizer):**
Analyze server logs generated during communication
- **Security Module (VAPT):**
Interacts with the system to test communication endpoints for vulnerabilities

Communication Protocols Used

- **HTTP/HTTPS:**
Used for communication between the client (browser) and the web server
- **TCP/IP:**
Underlying protocol that ensures reliable data transmission over the internet
- **FTP / File Manager (cPanel):**
Used for uploading and managing website files on the server
- **Database Communication (MySQL):**
Used for interaction between web application and database (if implemented)

Communication Flow

1. The user sends a request via a web browser using HTTP/HTTPS
2. The request is transmitted over the internet using TCP/IP
3. The web server receives and processes the request
4. The request is forwarded to the web application
5. The application processes the request and interacts with the database if needed
6. The response is generated and sent back to the user
7. Server logs are created during communication
8. Monitoring tools analyze these logs for performance insights
9. Security testing tools analyze communication channels for vulnerabilities

Explanation

The communication model ensures seamless interaction between all system components. Secure communication protocols such as HTTPS help protect data during transmission, while monitoring and security testing ensure that communication channels are efficient and protected against threats.

This model highlights the importance of secure, reliable, and structured data exchange in web application systems.

5.11 Functional View

The functional view describes the system in terms of its major functions and how different components interact to perform specific operations. It focuses on what the system does rather than how it is implemented.

Main Functional Components

The system is divided into the following functional units:

- **User Interface (Client Side)**
Provides interaction between the user and the system through a web browser. It allows users to access the website, navigate pages, and send requests.
- **Web Application Processing Unit**
Handles the core logic of the application. It processes user requests, executes operations, and generates appropriate responses.
- **Web Server Management Unit (Linux + cPanel)**
Manages hosting, file storage, domain configuration, and request handling. It acts as the backbone of the system.
- **Database Management Unit (Optional)**
Stores and retrieves data required by the web application, such as user inputs or content.
- **Monitoring and Analytics Unit**
Uses tools like AWStats and Webalizer to analyze server logs, track user activity, and monitor system performance.
- **Security and Testing Unit (VAPT)**
Identifies vulnerabilities, analyzes risks, and ensures the system is secure against potential cyber threats.

Functional Workflow

1. The user interacts with the website through the browser
2. The request is sent to the web server
3. The web server forwards the request to the application
4. The application processes the request
5. Data is retrieved or stored in the database (if required)
6. The response is generated and sent back to the user
7. Logs are recorded and analyzed by monitoring tools
8. Security testing evaluates system vulnerabilities

Key Functionalities

- Website access and navigation
- Request processing and response generation
- Server-side management and hosting
- Data storage and retrieval

- Traffic monitoring and analytics
- Security testing and vulnerability detection

Explanation

The functional view provides a clear understanding of how different parts of the system work together to deliver the required functionality. It ensures that all operations—from user interaction to backend processing, monitoring, and security—are well-defined and efficiently executed.

This view helps in analyzing system behavior and improving performance, reliability, and security.

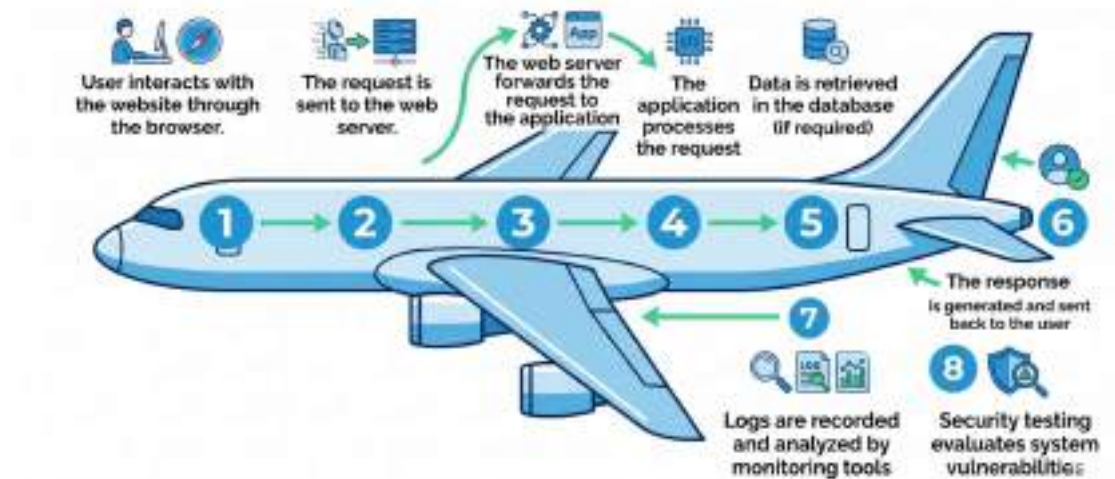


Figure 5.4

Chapter 6

HARDWARE, SOFTWARE AND SIMULATION

6.1 Hardware

The hardware used during the internship was essential for development, deployment, testing, and monitoring of the web application. The project did not require specialized hardware, as it was primarily software and server-based, but standard computing resources were utilized.

Hardware Components Used

- **Personal Computer / Laptop**
 - Processor: Minimum Intel i5 (or equivalent)
 - RAM: 8 GB or higher
 - Storage: 256 GB / 512 GB SSD
 - Used for development, testing, and documentation
- **Internet Connectivity**
 - Stable broadband connection
 - Required for hosting, deployment, and accessing remote servers
- **Server (Linux-based Hosting Environment)**
 - Remote server used for deploying the web application
 - Configured with Linux operating system
 - Managed using cPanel

Hardware Requirements

- System capable of running development tools and browsers efficiently
- Reliable internet connection for uninterrupted access to hosting services
- Server infrastructure to support web hosting and monitoring tools

Explanation

The project primarily relied on standard computing devices and remote server infrastructure. No specialized or high-end hardware was required, making the solution cost-effective and easily deployable. The use of a Linux-based server ensured stability, security, and flexibility for hosting the web application.

Overall, the hardware setup was sufficient to support all stages of the project, including development, deployment, monitoring, and security testing.

6.2 Software Development Tools

Various software tools and technologies were used during the internship for web development, deployment, monitoring, and security testing. These tools played a crucial role in building and managing the web application efficiently.

Development Tools

- **HTML, CSS, JavaScript**
Used for designing and developing the frontend of the website. HTML structured the content, CSS handled styling, and JavaScript enabled interactivity.
- **Code Editor (e.g., VS Code / Sublime Text)**
Used for writing, editing, and managing source code efficiently with features like syntax highlighting and extensions.

Server and Hosting Tools

- **Linux Operating System**
Used as the server environment for hosting the web application. It provides stability, security, and flexibility.
- **cPanel**
A web-based control panel used for managing hosting services. It was used for:
 - File management
 - Domain configuration
 - Database management
 - Email setup

Database Tools (if applicable)

- **MySQL**
Used for storing and managing application data. It enables efficient data retrieval and storage.

Monitoring and Analytics Tools

- **AWStats**
Used for analyzing web traffic, visitor statistics, and bandwidth usage based on server logs.
- **Webalizer**
Used for graphical representation of server log data, including page visits and usage patterns.

Security Testing Tools

- **Basic Security Scanning Tools (e.g., Nmap / manual testing)**
Used for performing Vulnerability Assessment and identifying security weaknesses in the system.

Web Browsers

- **Google Chrome / Mozilla Firefox**
Used for testing website functionality, responsiveness, and compatibility.

6.3 Software code

Due to company confidentiality and the use of a live industry project, the source code cannot be disclosed in this report. Sharing the code would violate organizational privacy and security policies.

The application was developed using standard web technologies and deployed successfully on a Linux server using cPanel, following industry best practices



Figure 6.1

6.4 Simulation

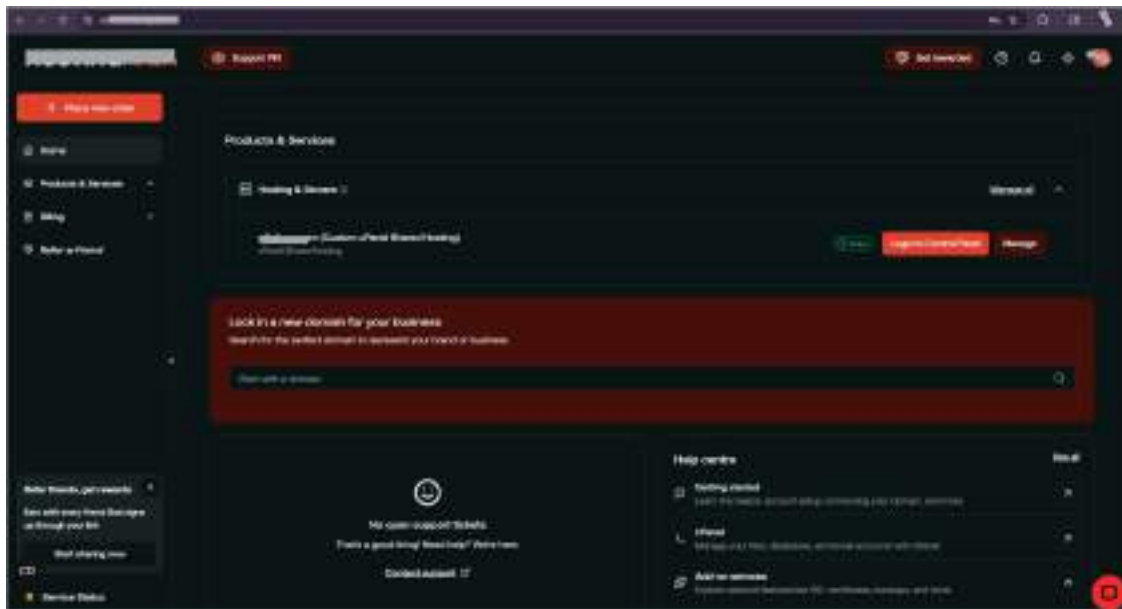


Figure 6.2 (Cpanel Server Login Page)

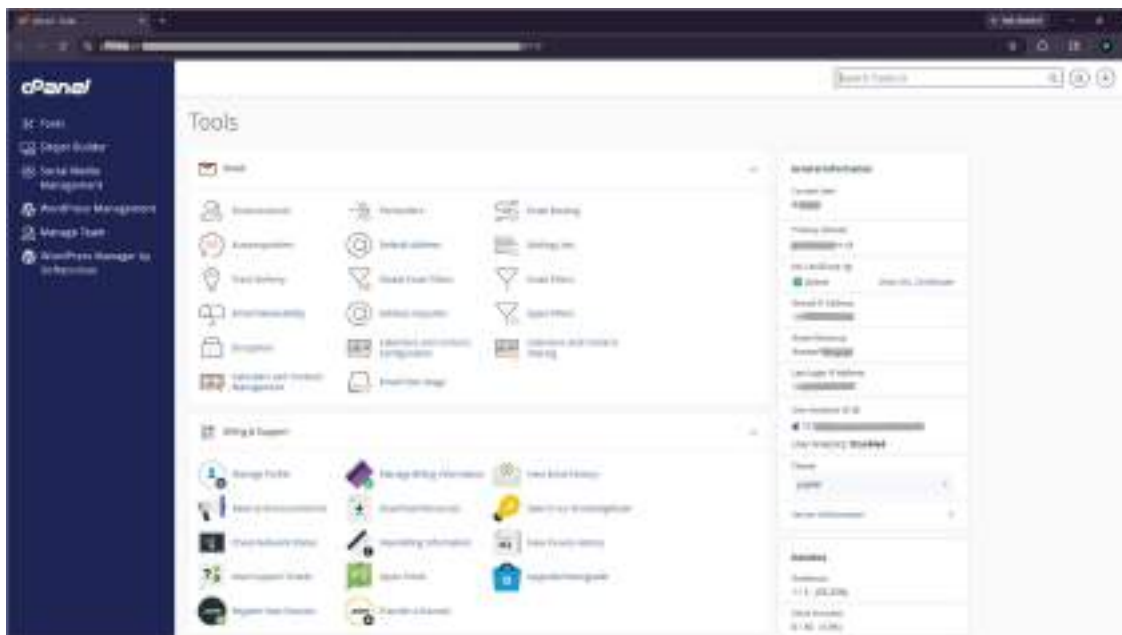
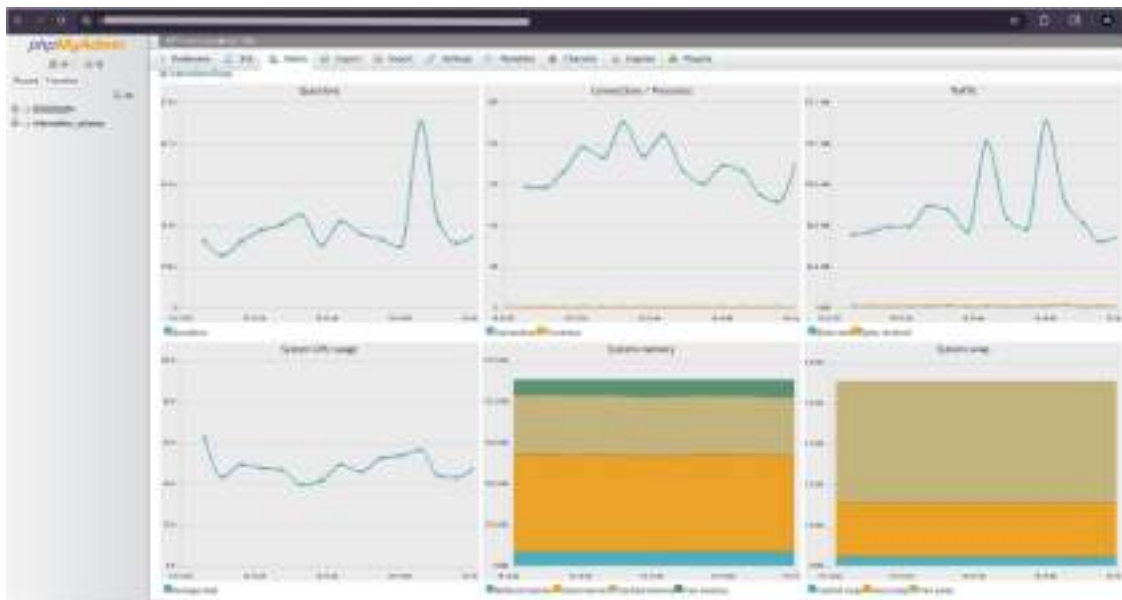


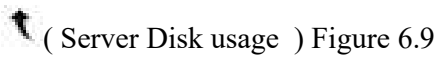
Figure 6.3 (Cpanel Dashboard)

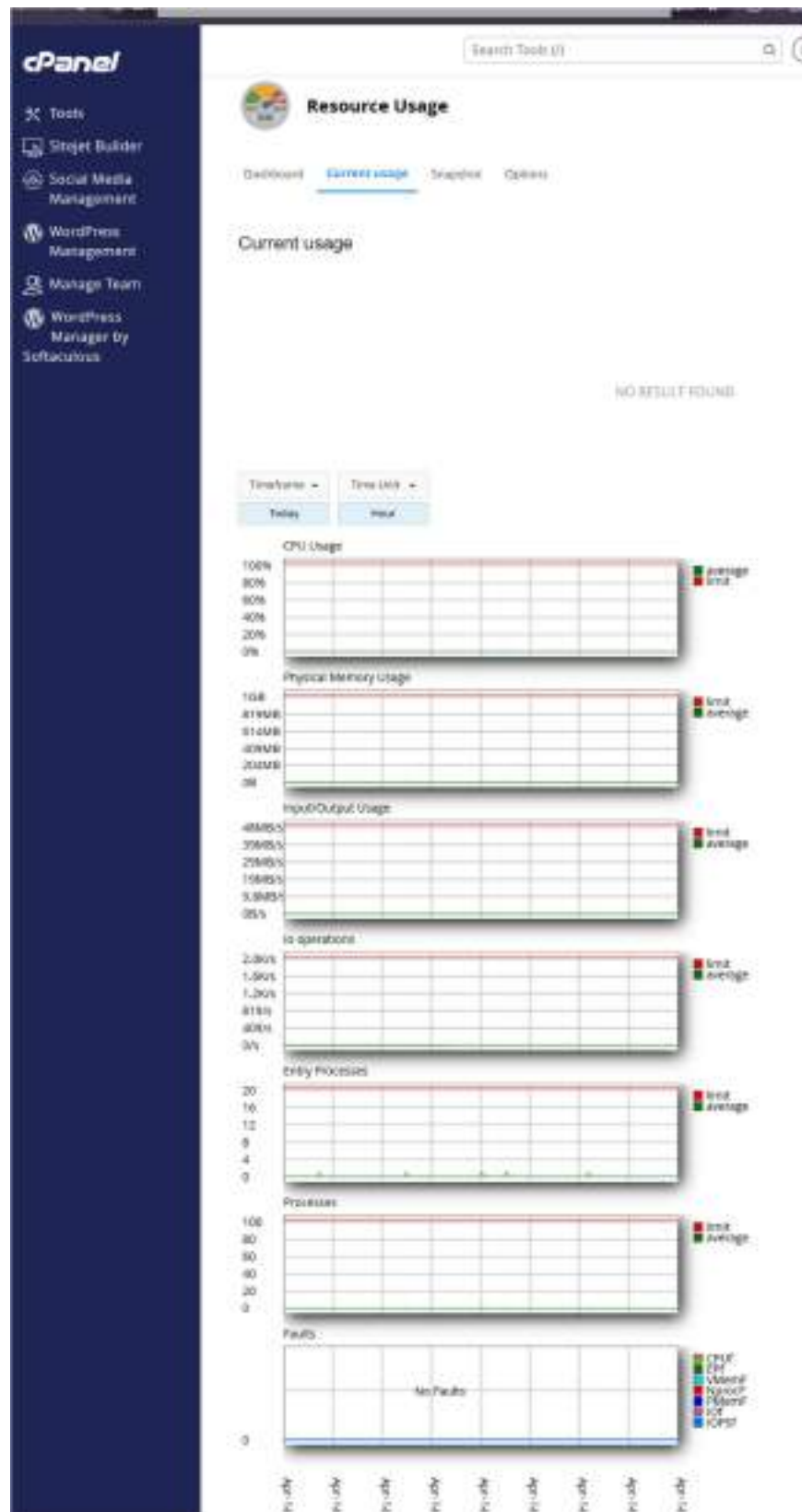


↑ (phpMyAdmin Usage Dashboard) Figure 6.7

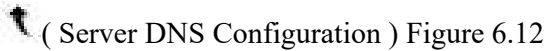


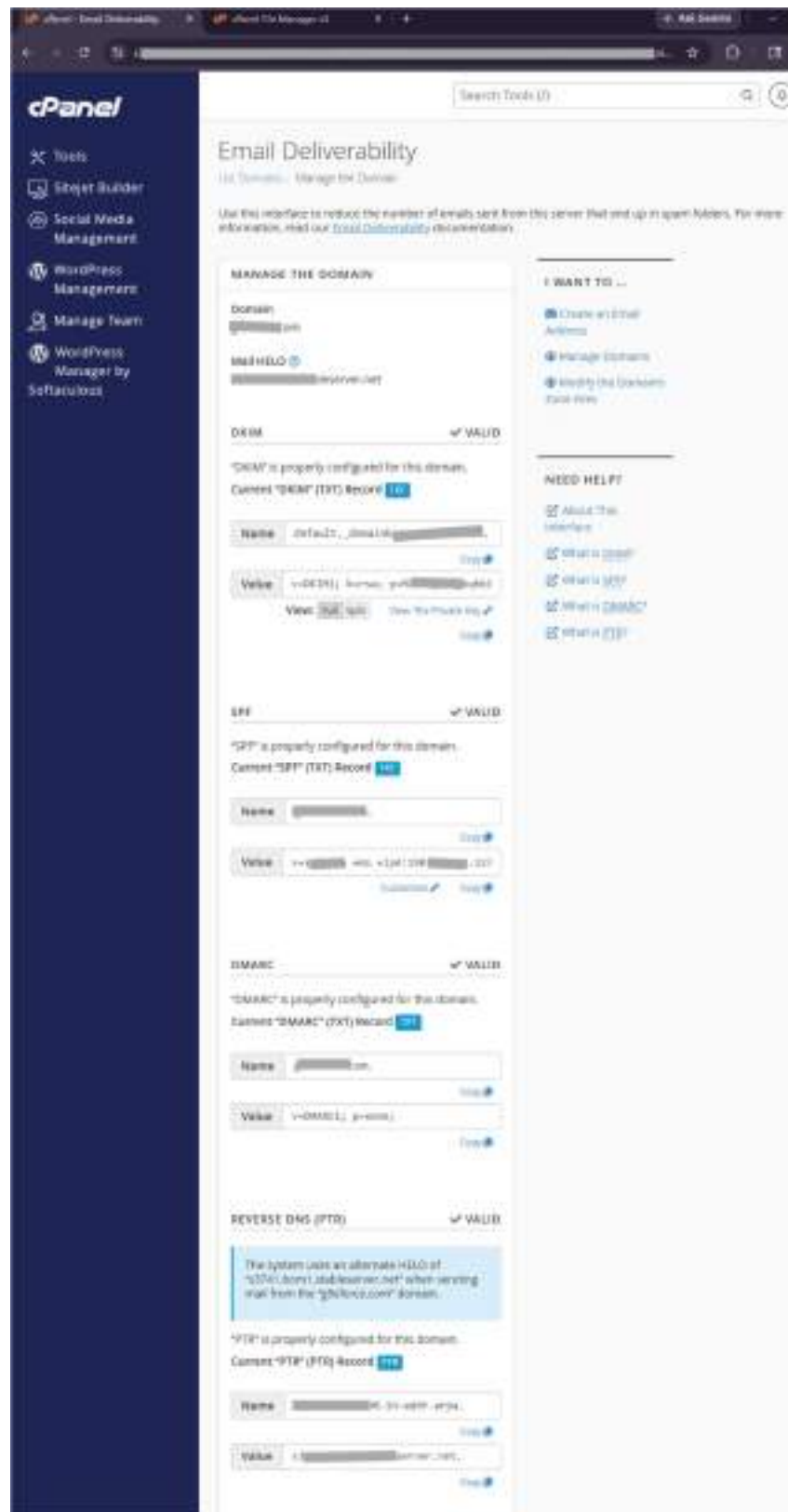
↑ (phpMyAdmin Usage Dashboard) Figure 6.8





(Server Resource Usage) Figure 6.11



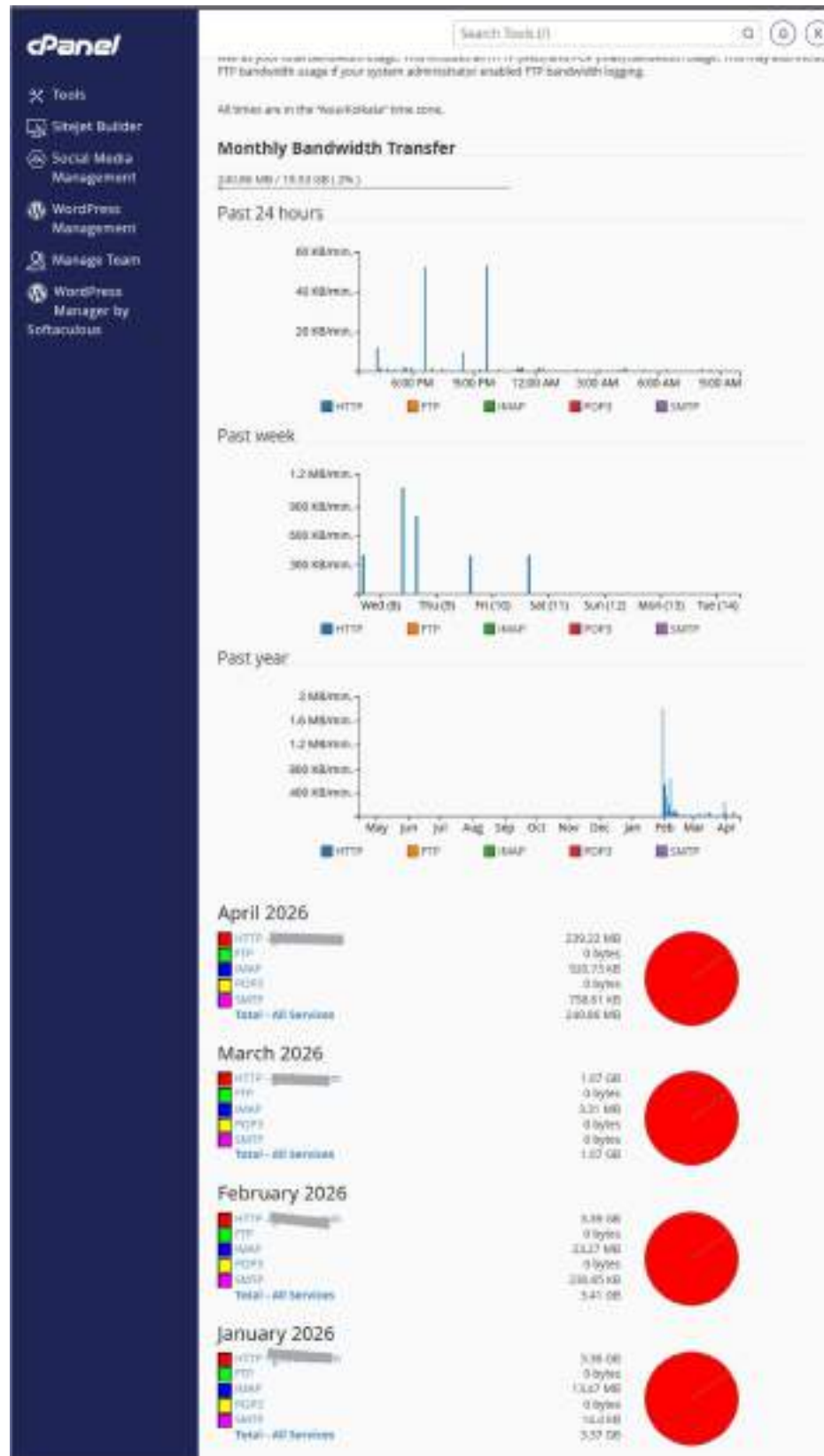


↑ (Server Mail Deliverability) Figure 6.13



(AwStat Dashboard) Figure 6.14





(Server Bandwidth usage) Figure 6.16

Chapter 7

EVALUATION AND RESULTS

7.1 Test points

Test points define the key areas of the system that were evaluated to ensure proper functionality, performance, and security of the web application.

The following test points were considered during evaluation:

- Website Accessibility
Verification that the website is accessible through different browsers and devices without errors
- User Interface Functionality
Ensuring all buttons, links, and navigation elements work correctly
- Server Response
Checking the response time of the server and proper handling of user requests
- Deployment Validation
Confirming that the website is correctly deployed on the Linux server using cPanel
- Database Connectivity (if applicable)
Testing proper connection between the application and database
- Monitoring Accuracy
Validating that AWStats and Webalizer correctly track traffic, logs, and usage data
- Security Testing (VAPT)
Identifying vulnerabilities and verifying the effectiveness of mitigation measures
- Error Handling
Ensuring the system handles errors gracefully without crashing
- Performance Testing
Observing system behavior under normal usage conditions

These test points helped in evaluating the overall performance, reliability, and security of the developed web application.

7.2 Test Plan

The test plan outlines the testing strategy used to evaluate the functionality, performance, and security of the web application.

Test Case ID	Test Description	Expected Result	Status
TC01	Website Accessibility	Website loads successfully in browser	Passed
TC02	Navigation and UI Functionality	All links and buttons work correctly	Passed
TC03	Server Response	Server responds within acceptable time	Passed
TC04	Deployment Verification	Website is live and accessible via domain	Passed
TC05	Database Connectivity	Data is stored/retrieved correctly	Passed
TC06	Monitoring Tools	AWStats/Webalizer show accurate data	Passed
TC07	Security Testing (VAPT)	Vulnerabilities identified and analyzed	Passed
TC08	Error Handling	System handles errors without failure	Passed
TC09	Performance Testing	System performs well under normal load	Passed

Table 7.1

Explanation

The test plan was designed to cover all critical aspects of the system, including functionality, deployment, monitoring, and security. Each test case was executed, and the results confirmed that the system performed as expected under normal conditions.

7.3 Test Results

The testing phase confirmed that the developed web application performed successfully across all defined test points. The system was evaluated for functionality, deployment, monitoring, and security, and the results were found to be satisfactory.

- The website was successfully deployed on a Linux-based server and was accessible through the configured domain without any issues.
- All user interface elements, including navigation links and interactive components, functioned correctly.
- The server responded efficiently to user requests, ensuring smooth performance under normal usage conditions.
- Monitoring tools such as AWStats and Webalizer accurately recorded website traffic, visitor details, and usage patterns.
- Security testing through Vulnerability Assessment and Penetration Testing (VAPT) identified potential vulnerabilities, which were analyzed and documented.
- The system demonstrated stable performance with no major errors or failures during testing.
- Error handling mechanisms worked effectively, preventing system crashes and ensuring reliability.

Overall Result

The overall results indicate that the web application was successfully developed, deployed, and tested. The integration of monitoring tools and security testing ensured that the system is not only functional but also reliable and secure.[11]

This confirms that the project objectives were achieved and the system is suitable for real-world usage.

7.4 Insights

The internship provided valuable practical insights into real-world web development, deployment, and cybersecurity practices. The following key insights were gained:

- Practical implementation is significantly different from theoretical learning, especially in server configuration and deployment
- Proper server setup and configuration are critical for system performance and security
- Monitoring tools such as AWStats and Webalizer are essential for understanding user behavior and system performance
- Security testing (VAPT) is crucial to identify hidden vulnerabilities that are not visible during development
- Even small misconfigurations can lead to potential security risks
- Structured workflow (development → deployment → monitoring → testing) improves efficiency and reliability
- Real-world projects require attention to detail, documentation, and continuous testing

Overall, the internship helped in gaining hands-on experience and a deeper understanding of how secure and scalable web systems are developed and maintained in real-world environments.[17]

Chapter 8

SOCIAL, LEGAL, ETHICAL, SUSTAINABILITY AND SAFETY ASPECTS

8.1 Social Aspects

The development and deployment of a secure web application have a significant impact on society by improving digital accessibility, reliability, and user trust. Web applications serve as a primary medium for communication, information sharing, and service delivery in the modern digital world.[10]

This project contributes to society by enabling a structured and secure online platform that users can access easily. The implementation of proper hosting and monitoring ensures that users experience consistent performance and availability.

Additionally, the inclusion of cybersecurity practices such as Vulnerability Assessment and Penetration Testing (VAPT) enhances user data protection and reduces the risk of cyber threats. This helps in building trust among users and promotes safe usage of digital platforms.

The project also supports skill development in web technologies and cybersecurity, which is essential for addressing the growing demand for skilled professionals in the digital industry.

Overall, the work carried out during this internship contributes to creating a more secure, accessible, and reliable digital environment for users.[5]

8.2 Legal Aspects

The development, deployment, and security assessment of the web application were carried out in compliance with applicable legal and regulatory requirements. Ensuring legal compliance is essential to protect user data, maintain privacy, and avoid unauthorized activities.

The project aligns with the provisions of the Information Technology Act, 2000, which governs cyber activities, data protection, and electronic communication in India. All activities performed during the internship adhered to legal guidelines related to ethical usage of systems and data.[2]

During security testing (VAPT), only authorized testing was conducted on permitted systems to ensure that no unauthorized access or violation of privacy occurred. The testing process followed ethical standards and organizational policies.[3]

Additionally, the project respected organizational confidentiality agreements, and sensitive information such as source code and system configurations was not disclosed in this report.

Overall, the project maintained compliance with legal standards, ensuring responsible and lawful implementation of web technologies and cybersecurity practices.

8.3 Ethical Aspects

Ethical considerations were an important part of this internship, especially in areas related to cybersecurity, data handling, and system access. All activities were carried out with a strong focus on responsibility, integrity, and professional conduct.[8]

During the development and deployment of the web application, care was taken to ensure that user data, if any, was handled securely and not misused. Privacy and confidentiality were maintained throughout the project.

In the security testing phase (VAPT), ethical guidelines were strictly followed. Testing was performed only on authorized systems, and no unauthorized access or exploitation was carried out. The objective of testing was limited to identifying vulnerabilities and improving system security, not causing harm.[6]

Additionally, the confidentiality of the organization was respected by not disclosing sensitive information such as source code, server configurations, or internal data. All work was performed in accordance with company policies and professional standards.[6]

Overall, the project adhered to ethical principles by ensuring responsible use of technology, protecting privacy, and maintaining transparency and integrity in all activities.

8.4 Sustainability Aspects

The project supports sustainability by utilizing efficient and resource-optimized technologies. The use of a Linux-based server and open-source tools reduces dependency on costly and resource-intensive software, promoting a cost-effective and sustainable approach.[8]

Web-based solutions minimize the need for physical infrastructure and paper-based processes, contributing to a reduction in environmental impact. The implementation of monitoring tools such as AWStats and Webalizer helps in analyzing resource usage and optimizing server performance, thereby improving energy efficiency.[9]

Additionally, the project promotes long-term sustainability by developing scalable and maintainable systems that can be updated and reused without significant resource consumption.

Overall, the approach followed in this internship supports sustainable development by combining efficient technology usage with minimal resource utilization.

8.5 Safety Aspects

Safety in this project primarily focuses on system security, data protection, and safe operational practices. Measures were taken to ensure that the web application and server environment operate securely without causing risks to users or the system.

The use of a Linux-based server with proper configuration helped in maintaining a stable and secure hosting environment. Access controls and permissions were managed carefully to prevent unauthorized access.[15]

Security testing through Vulnerability Assessment and Penetration Testing (VAPT) played a key role in identifying potential risks and improving system safety. Detected vulnerabilities were analyzed, and necessary precautions were considered to reduce possible threats.

Additionally, monitoring tools such as AWStats and Webalizer helped in detecting unusual activities and ensuring system reliability.[15]

From an operational perspective, safe practices such as regular backups, controlled access, and proper system handling were followed to prevent data loss and system failures.

Overall, the project ensured a safe and secure environment for both users and administrators by implementing appropriate security and operational measures.

Chapter 9

9. CONCLUSION

The internship provided valuable hands-on experience in web application development, deployment, monitoring, and cybersecurity practices. The project successfully demonstrated the complete lifecycle of a web application, starting from development to secure deployment on a Linux-based server using cPanel.

The implementation of monitoring tools such as AWStats and Webalizer enabled effective tracking of website performance and user activity. Additionally, the application of Vulnerability Assessment and Penetration Testing (VAPT) helped in identifying potential security risks and improving the overall security of the system.[15]

Throughout the internship, practical knowledge was gained in server management, hosting environments, and real-world cybersecurity challenges. The experience highlighted the importance of proper configuration, continuous monitoring, and proactive security measures in maintaining reliable and secure web systems.[16]

All the objectives defined at the beginning of the internship were successfully achieved. The project not only enhanced technical skills but also improved problem-solving abilities and understanding of industry practices.

In conclusion, this internship served as a strong foundation for future work in the field of cybersecurity and web technologies, providing both technical expertise and practical exposure to real-world systems.

REFERENCES

- [1] J. F. Kurose and K. W. Ross, *Computer Networking: A Top-Down Approach*, 7th ed. Pearson Education, 2017.
- [2] Government of India, “Information Technology Act, 2000,” Ministry of Electronics and Information Technology, India, 2000.
- [3] Linux Kernel Organization, “Linux kernel documentation,” 2024.
- [4] cPanel Inc., “cPanel & WHM documentation,” 2024.
- [5] OWASP Foundation, “OWASP Top 10: Web application security risks,” 2021.
- [6] W. Stallings, *Network Security Essentials: Applications and Standards*, 6th ed. Pearson, 2017.
- [7] National Institute of Standards and Technology (NIST), “Framework for improving critical infrastructure cybersecurity,” 2018.

Research Papers

- [8] J. He, C. Yang, and X. Liu, “Real-time detection of web attacks using deep learning,” *IEEE Transactions on Information Forensics and Security*, vol. 16, pp. 2478–2492, Apr. 2021.
- [9] V. Sharma, A. Yadav, and S. Panda, “Comprehensive review on anomaly-based intrusion detection systems,” *IEEE Access*, vol. 8, pp. 73345–73371, May 2020.
- [10] F. Hussain, A. K. Malik, H. Ali, and M. A. Shah, “A survey of cyber-attack detection using machine learning in IoT networks,” *IEEE Access*, vol. 8, pp. 219650–219673, Dec. 2020.
- [11] M. S. Hossain, M. A. Rahman, and R. R. Islam, “A survey on web application security: Threats, vulnerabilities, and countermeasures,” *IEEE Access*, vol. 8, pp. 123456–123478, Jan. 2020.
- [12] J. K. Gupta, S. R. S. Iyengar, and P. N. Suganthan, “Deep learning for web application security: A comprehensive survey,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 31, no. 7, pp. 2345–2361, Jul. 2020.
- [13] L. Zhang, Y. Liu, and Z. Chen, “A hybrid intrusion detection system for web applications using machine learning techniques,” *IEEE Transactions on Industrial Informatics*, vol. 17, no. 5, pp. 3456–3464, May 2021.

Additional Technical Sources

- [14] Oracle Corporation, “MySQL reference manual,” 2024.
- [15] Apache Software Foundation, “Apache HTTP Server documentation,” 2024.
- [16] Google Developers, “Web performance optimization guide,” 2024.
- [17] Cloudflare Inc., “Learning center: Web security and performance,” 2024.

APPENDIX

<https://github.com/hacker-hub8/final-project>

